

Early Disease Prediction System

*Diabetes and heart-disease risk before symptoms —
with honest uncertainty, an independent second opinion, and a preventive plan*

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Live application: **healthpredictor092005.streamlit.app**
Source code: github.com/2005Maverick/early-disease-prediction-

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Abstract

This report presents the **Early Disease Prediction System**, a machine-learning application that estimates a patient's risk of diabetes or heart disease from routine lifestyle, demographic and clinical measurements, before symptoms appear. The system is designed around a principle that most student and industry prototypes ignore: *a prediction is only useful if the user can tell how much to trust it*. Instead of a single risk number, an *Uncertainty Engine* — 200 gradient-boosting pipelines, each trained on a different bootstrap resample of the study population — produces a full risk *distribution* per patient, whose spread is an honest measure of confidence. A second, entirely model-free method (*Patients Like You*, a 50-nearest-neighbour lookup over the standardized study population) provides an independent second opinion, and agreement between the two methods is surfaced to the user. All reported metrics are strictly out-of-fold (5-fold stratified cross-validation): the deployed ensembles achieve a ROC AUC of **83.5%** for diabetes (Pima study, 768 patients) and **90.0%** for heart disease (Cleveland study, 303 patients) at a recall of at least 85% by construction, the alert threshold being derived from an explicit clinical rule rather than tuned by hand. Empirically, the uncertainty is informative: the error rate rises from 6% among the quarter of patients with the narrowest confidence bands to 43% among the widest (diabetes; 4% to 46% for heart disease) — the system measurably knows when it does not know. The delivery layer is a deployed Streamlit application styled as an interactive lab report, with per-patient risk drivers, a what-if lifestyle simulator, and a guideline-based preventive plan. The architecture is disease-agnostic: each disease is a plug-in configuration file, so extending the system to a third disease requires one new file and a retraining run.

Keywords: early disease prediction, uncertainty quantification, bootstrap ensembles, gradient boosting, nearest neighbours, calibration, explainable AI, Streamlit.

1. Introduction

Type-2 diabetes and cardiovascular disease are two of the most common chronic conditions worldwide, and both share a property that makes them natural targets for machine learning: they develop silently for years before diagnosis, while leaving statistical fingerprints in measurements that are cheap and routine — blood glucose, body-mass index, blood pressure, cholesterol, age, family history. Landmark trials such as the Diabetes Prevention Program have shown that when elevated risk is identified early, lifestyle interventions can cut disease incidence dramatically (58% for diabetes) [8]. The value of an early-warning model is therefore not the prediction itself but the *action* it enables.

Most classroom risk-prediction projects follow a fixed template: train a classifier on a public dataset, report accuracy on data the model has partially seen, and wrap the result in a web form. This project deliberately departs from that template in four ways. First, it reports **only out-of-fold metrics** — every number in this report was measured on patients the model never saw during training. Second, it treats **uncertainty as a first-class output**: the user sees not just "84.8% risk" but how much 200 independently trained models disagree about that number. Third, it provides an **independent second opinion** computed without any model at all, so that trust does not rest on a single method. Fourth, it converts predictions into **concrete preventive actions** traced to clinical guidelines, closing the loop from data to decision.

The result is a working, publicly deployed system covering two diseases, built on one shared engine, with a test suite, a reproducible training pipeline, and a figure-generation script that produced every chart in this report from the actual artifacts.

2. Literature Review

2.1 Tabular machine learning

Despite a decade of deep-learning progress, tree-based ensembles remain the strongest general-purpose learners for small and medium tabular datasets. Grinsztajn et al. [5] show systematically why tree ensembles outperform neural networks on tabular data of this scale, attributing the gap to the rotation-variance of trees and their robustness to uninformative features. Gradient boosting [3] builds an additive sequence of shallow trees, each correcting the residual errors of its predecessors, and is the base learner used here. Random forests [2] and regularized logistic regression serve as baselines; with 768 and 303 patients respectively, our datasets sit squarely in the regime where these classical methods are expected to dominate, and our own comparison (Section 7.1) confirms it.

2.2 Uncertainty quantification

A single classifier's probability output conflates two different things: the risk of the patient, and the model's confidence in that estimate. The bootstrap [1, 9] offers a conceptually simple separation: train many models on resamples of the data, and read the spread of their predictions as sampling uncertainty. Bagging [1] originally used this for variance reduction; we retain the averaged prediction for accuracy but additionally expose the full distribution to the user. This is closely related to deep ensembles in the neural network literature, widely regarded as the most reliable practical uncertainty method, here applied in its natural tabular form.

2.3 Explainability and case-based reasoning

Post-hoc explanation methods such as SHAP assign each feature a contribution to a prediction. We use a deliberately simpler scheme — median substitution: replace one of the patient's values with the population's typical value, re-run the model, and report the risk change. It is model-agnostic, exact for the deployed ensemble, and explainable to a non-technical audience in one sentence. The *Patients Like You* view draws on the oldest idea in pattern recognition, nearest-neighbour classification [7]: presenting the outcomes of the fifty most similar real patients is both an independent risk estimate and a form of case-based explanation that clinicians find natural. Finally, calibration — whether a stated "70%" corresponds to a 70% observed event rate — is assessed with reliability diagrams as in the classical calibration literature [10].

3. Problem Statement

Can routine lifestyle, demographic and clinical measurements be combined into a per-patient disease-risk assessment that is (i) measured honestly, on patients the model has never seen; (ii) accompanied by a truthful statement of the model's own confidence; (iii) corroborated by an independent, model-free method; and (iv) actionable, in the form of specific preventive steps traced to clinical guidelines — all delivered in an interface a non-technical user can operate?

4. Proposed Solution

The system is organised in two phases (Figure 1). Offline, a training pipeline loads each disease's dataset, repairs its hidden missing values, evaluates four candidate models with stratified 5-fold cross-validation, derives the alert threshold from a stated recall rule, and saves the deployed artifacts. Online, a gated intake form collects the patient's values; on explicit submission the trained ensemble scores the patient *once*, the 50 most similar real patients are retrieved, and a five-section assessment report is rendered. Nothing is predicted until the user asks — the interface enforces a clear causal moment between input and output.

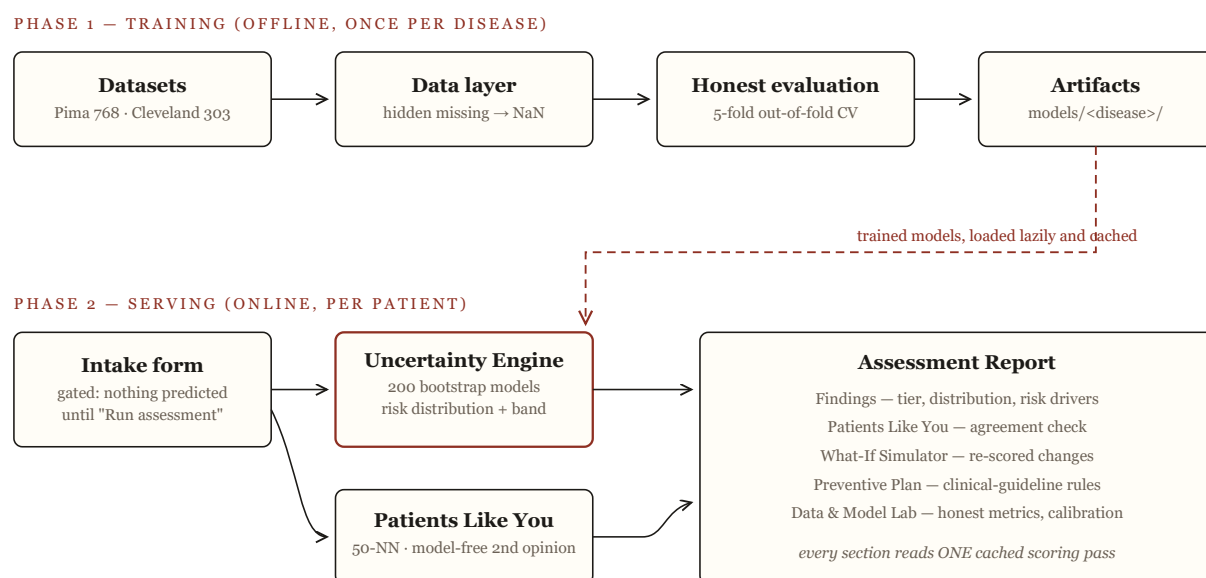


Figure 1: System architecture: the offline training phase (top) and the online per-patient serving phase (bottom).

A key architectural decision is that the engine is **disease-agnostic** (Figure 2). Everything specific to a disease — dataset location, feature list and semantics, which values are physiologically impossible, the intake form specification, the what-if scenarios and the prevention rules — lives in one configuration object per disease. The ensemble, the neighbour engine, the threshold rule, the explanation methods, the training pipeline and the entire user interface are shared. Supporting a third disease means writing one new configuration file and running the trainer.

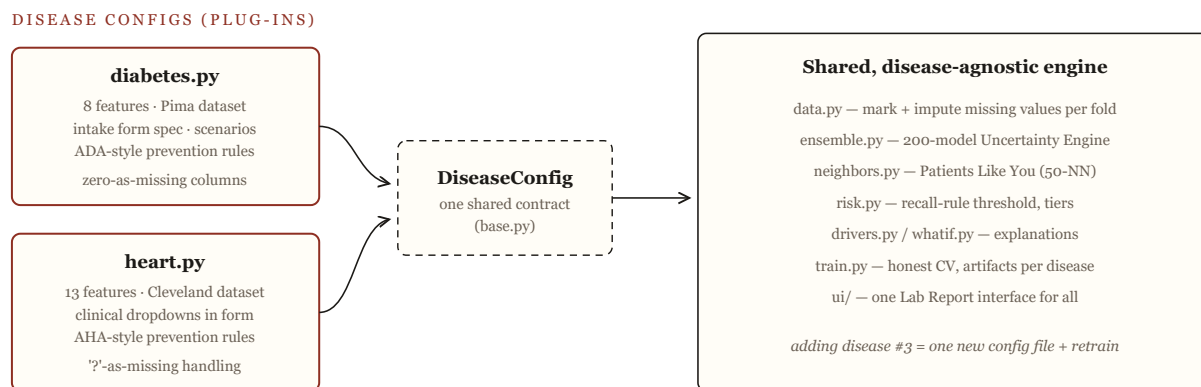


Figure 2: The disease registry: diseases plug into a shared engine through a single configuration contract.

5. Data

5.1 Datasets

Two classical, publicly available clinical datasets are used (Table 1). The Pima Indians Diabetes Database [6] records eight measurements for 768 women aged 21 and above, with a binary outcome indicating development of diabetes. The Cleveland Heart Disease dataset [4] records thirteen clinical features for 303 patients undergoing angiography; the original five-level diagnosis is binarized (any vessel narrowing > 50% counts as disease), which is the standard treatment in the literature.

Table 1: Datasets used in this project.

Disease	Source study	Patients	Features	Positive rate
Diabetes	Pima (NIDDK) [6]	768	8	34.9%
Heart disease	Cleveland Clinic (UCI) [4]	303	13	45.9%

5.2 The hidden-missing-values problem

The Pima dataset contains a famous trap that most projects on it silently fall into: zeros in Glucose, BloodPressure, SkinThickness, Insulin and BMI. A blood-sugar or BMI of zero is physiologically impossible — these are undocumented *missing values*, not measurements. Feeding them to a model as real numbers distorts every downstream statistic. Our data layer marks them as missing (Figure 4); imputation with the training-fold median then happens *inside* each model pipeline, so it is re-fitted on every cross-validation fold and no information can leak from test to train. The Cleveland dataset marks its six missing entries explicitly with "?"; the same machinery absorbs them. The same mechanism also serves the user interface: a patient who does not know their serum insulin can mark it "unknown", and the model imputes it transparently.

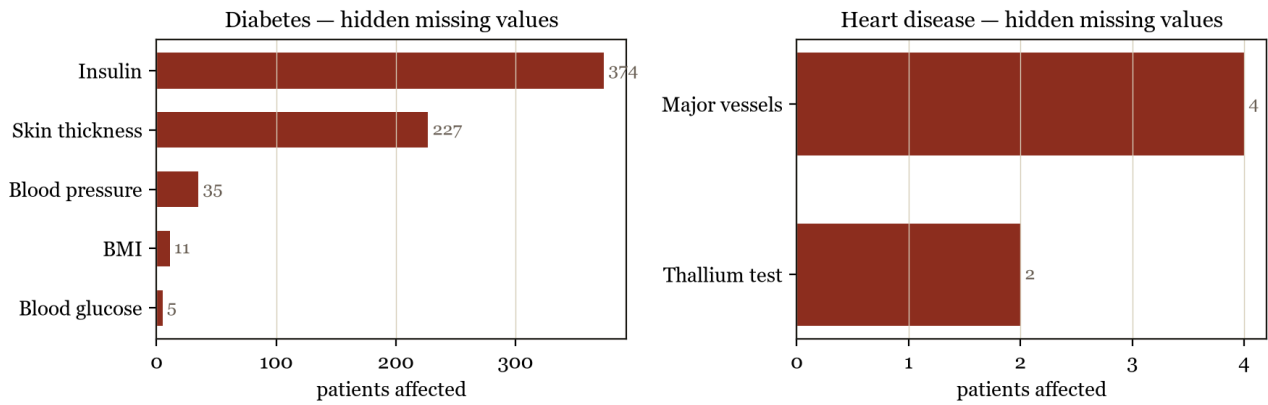


Figure 3: Hidden missing values per feature. In the Pima data, nearly half of all patients (374 of 768) have a physiologically impossible zero recorded for insulin.

5.3 Exploratory analysis

Figures 5 and 6 show the distributions of headline features split by outcome. The separations visible to the naked eye — glucose and BMI for diabetes; maximum achieved heart rate, ST depression and age for heart disease — anticipate the feature importances the models later learn, and match clinical intuition.

Diabetes: feature distributions by outcome

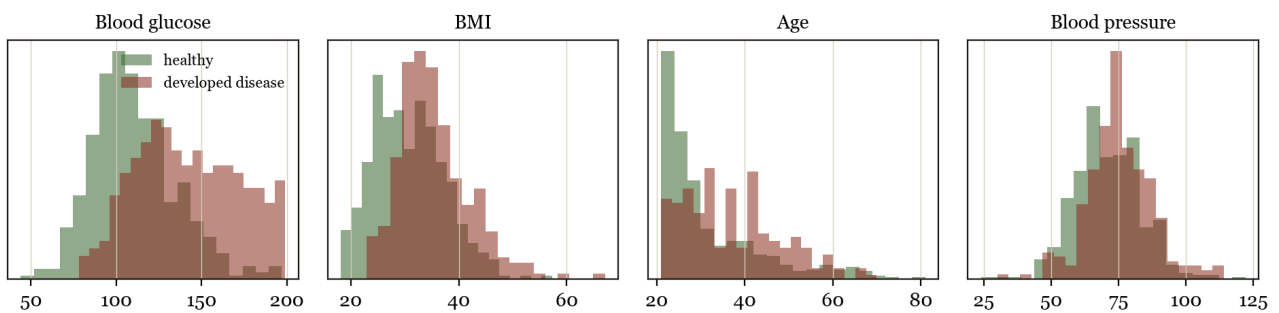


Figure 4: Diabetes: feature distributions for patients who stayed healthy (green) versus developed the disease (red).

Heart disease: feature distributions by outcome

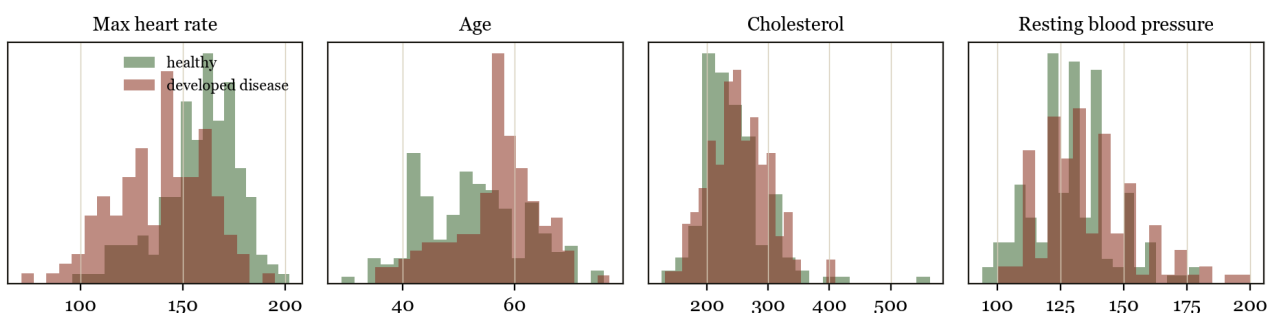
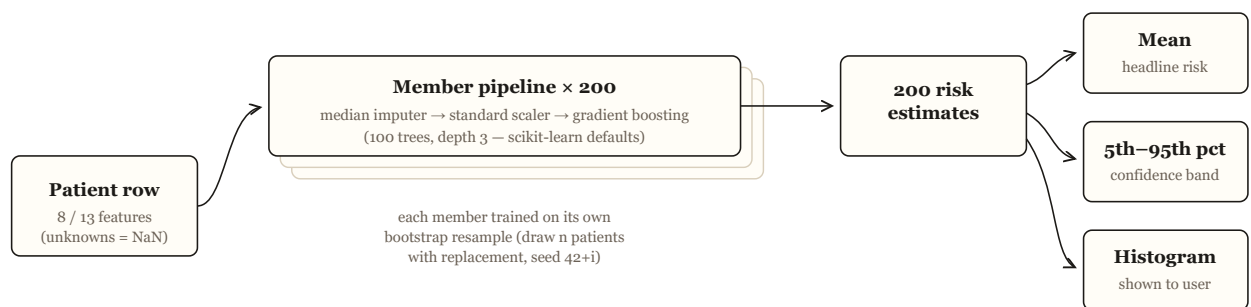


Figure 5: Heart disease: the same comparison for the Cleveland features.

6. Methodology

6.1 The Uncertainty Engine

The deployed model is a bootstrap ensemble of 200 identical scikit-learn pipelines (Figure 3, Table 2). Each member consists of a median imputer, a standard scaler and a gradient-boosting classifier with library-default hyperparameters (100 trees of depth 3, learning rate 0.1). Each member is trained on its own bootstrap resample — n patients drawn with replacement from the n available — with a distinct random seed. At prediction time every member outputs a probability, giving a 200-value distribution per patient: its mean is the headline risk score, its 5th–95th percentile range is reported as a 90% confidence band, and the full histogram is shown to the user.



*"We ask 200 slightly different models and plot all their answers.
Agreement means confidence; spread means the system honestly is not sure."*

Figure 6: Inside the Uncertainty Engine: 200 bootstrap-trained pipelines produce a risk distribution rather than a single number.

Table 2: Deployed model configuration.

Component	Choice	Rationale
Imputation	median, per training fold	robust; no leakage
Scaling	standardization	method-agnostic pipeline
Base learner	gradient boosting (100 trees, depth 3, lr 0.1)	strongest single model class for tabular data [5]
Ensemble	200 bootstrap members, seeds 42–241	uncertainty via bootstrap [1, 9]; variance reduction
Hyperparameters	scikit-learn defaults, untuned	with ~600 training rows per fold, aggressive tuning fits noise; evaluation integrity was prioritised

Why not deep learning? With 768 and 303 patients, the datasets are two to three orders of magnitude below the regime where neural approaches become competitive on tabular data [5]. The novelty of this system is deliberately *not* an exotic architecture but what is built around a proven one: uncertainty, corroboration, honest evaluation and actionable explanation.

6.2 Patients Like You

The second opinion uses no model at all. All study patients are imputed and standardized once; a nearest-neighbour index [7] then retrieves the 50 patients closest to the query in Euclidean distance, and the system reports how many of them actually developed the disease. Because this estimate shares no machinery with the ensemble, agreement between the two is meaningful evidence, and disagreement flags an unusual patient — exactly the case where extra caution is warranted. The interface states explicitly which situation applies.

6.3 The decision threshold as a stated rule

Converting a risk score into an alert requires a threshold, and in a medical screening context the costs of the two error types are asymmetric: missing a true future patient costs more than a false alarm. The threshold is therefore derived from an explicit rule — **among all thresholds whose out-of-fold recall is at least 85%, choose the one with the highest precision** — rather than defaulting to 0.5 or tuning by eye. The rule yields 23.8% for diabetes and 27.3% for heart disease (Figure 10). Both the rule and its consequence are displayed in the application itself.

6.4 Explanations without jargon

Per-patient explanations use median substitution: for each feature, the patient's value is replaced by the population median and the ensemble is re-run; the resulting risk drop is that feature's personal contribution ("your glucose alone adds 34 points"). The what-if simulator applies the same mechanism prospectively, re-scoring the patient under achievable changes (lose weight, lower glucose or cholesterol) that are generated per disease and only when applicable to the individual. Preventive recommendations are deliberately *rule-based*, written from published clinical reference ranges (e.g. impaired glucose tolerance at 140–199 mg/dL; stage-2 hypertension at ≥ 140 mm Hg), because prevention advice must be traceable to a guideline, not to a statistical artifact.

6.5 Evaluation protocol

All models are evaluated with stratified 5-fold cross-validation. Within each fold, the full pipeline — imputation, scaling, and for the ensemble the entire 40-member bootstrap procedure — is fitted from scratch on the training portion and scored on the held-out portion, so every prediction used in this report is *out-of-fold*: made by a model that never saw that patient. The deployed 200-member ensembles are then trained on all data. The threshold rule, the calibration analysis and the uncertainty analysis below all operate exclusively on the out-of-fold scores. The unit-test suite (15 tests) covers the data repair, threshold rule, tier logic, explanation and scenario builders, and both engines.

7. Results

7.1 Model comparison

Tables 3 and 4 compare the Uncertainty Ensemble against three single-model baselines on identical folds, all evaluated at each disease's rule-derived threshold.

Table 3: Diabetes — out-of-fold performance (% , threshold 23.8).

Model	Accuracy	Precision	Recall	F1	ROC AUC
Uncertainty Ensemble	73.1	57.7	85.1	68.8	83.5
Logistic Regression	70.4	55.0	84.7	66.7	83.5
Random Forest	69.9	54.2	88.4	67.2	83.3
Gradient Boosting	73.1	57.9	83.2	68.3	82.6

Table 4: Heart disease — out-of-fold performance (% , threshold 27.3).

Model	Accuracy	Precision	Recall	F1	ROC AUC
Uncertainty Ensemble	78.6	72.6	85.6	78.6	90.0
Logistic Regression	80.9	74.9	87.8	80.8	90.9
Random Forest	74.9	66.3	92.1	77.1	91.2
Gradient Boosting	80.2	74.5	86.3	80.0	88.5

On diabetes the ensemble matches or beats every baseline on every aggregate metric. On the smaller Cleveland dataset the picture is more nuanced — logistic regression and random forest edge slightly ahead on AUC (90.9 and 91.2 versus 90.0). We report this openly: the ensemble was nevertheless chosen for deployment because it is the only candidate that produces a per-patient uncertainty distribution, and Section 7.4 shows that this uncertainty carries real information a point estimate cannot. Within roughly one AUC point, the honest confidence band is worth more than the marginal discrimination.

7.2 Discrimination and operating point

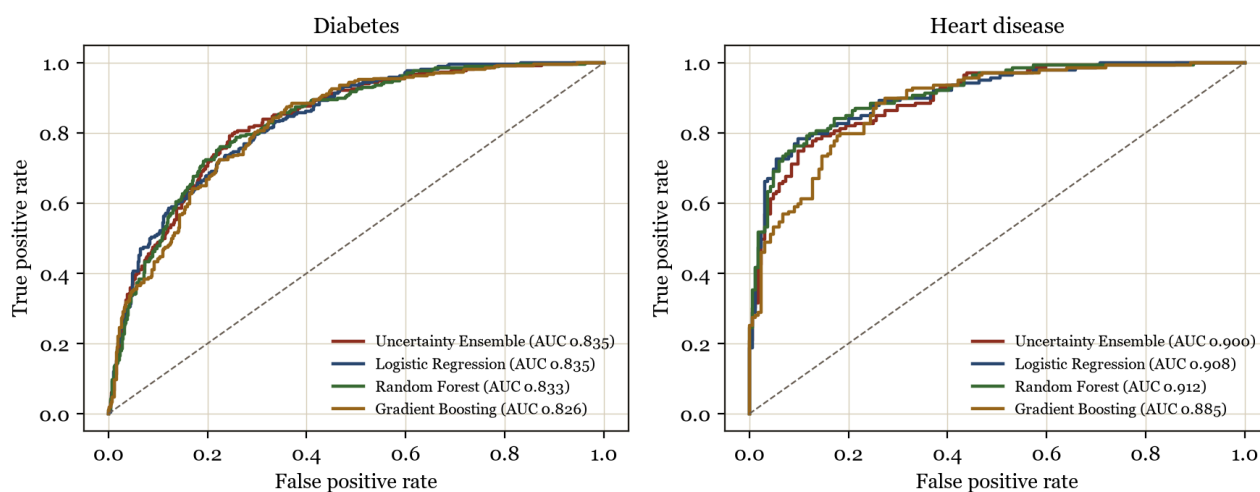


Figure 7: ROC curves for all four models on both diseases (out-of-fold).

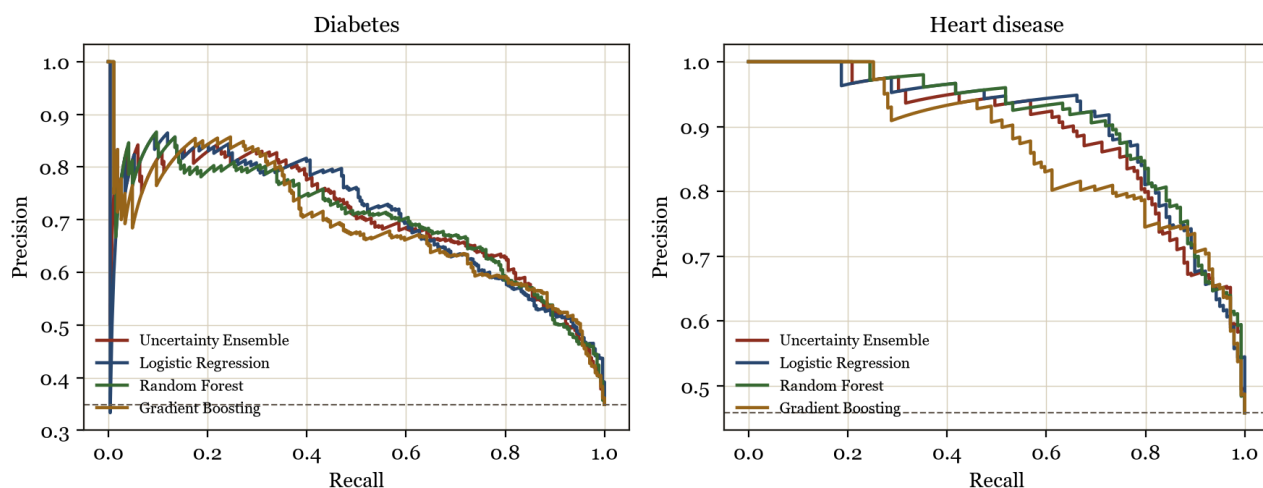


Figure 8: Precision–recall curves; the dashed line marks each disease's positive-class base rate.

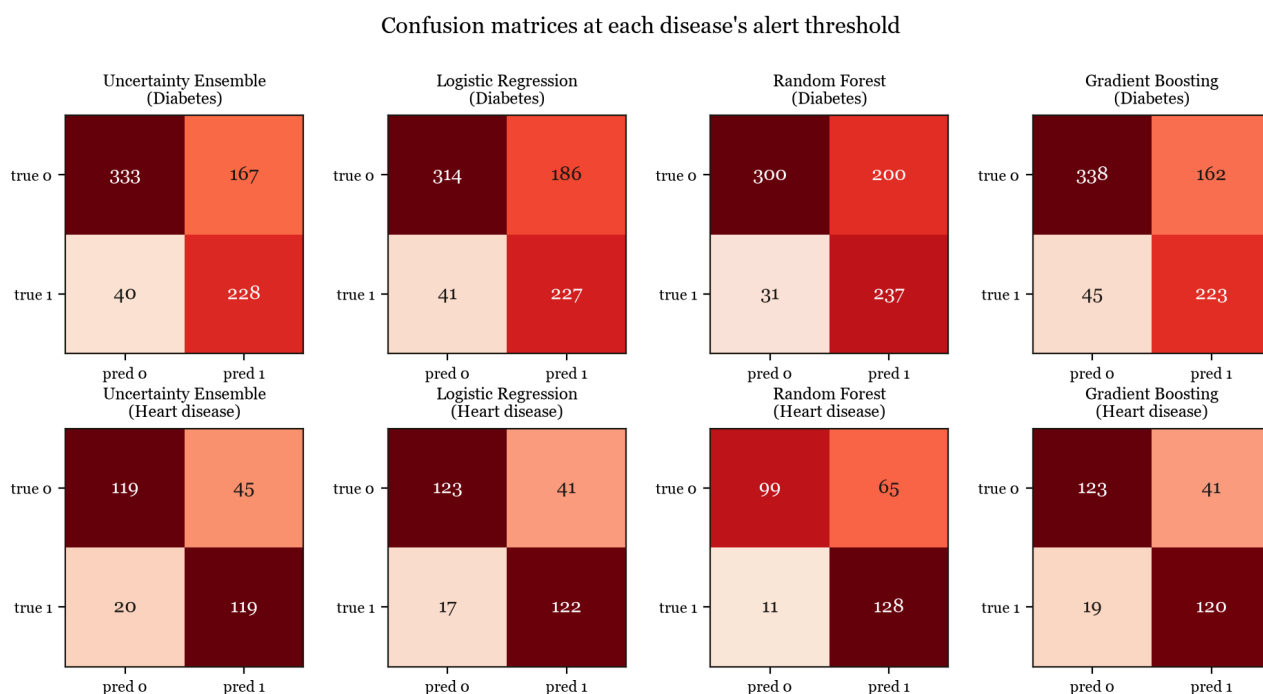


Figure 9: Confusion matrices at each disease's alert threshold. The recall-first rule deliberately trades false positives for very few missed patients.

The alert threshold is a stated rule, not a magic number

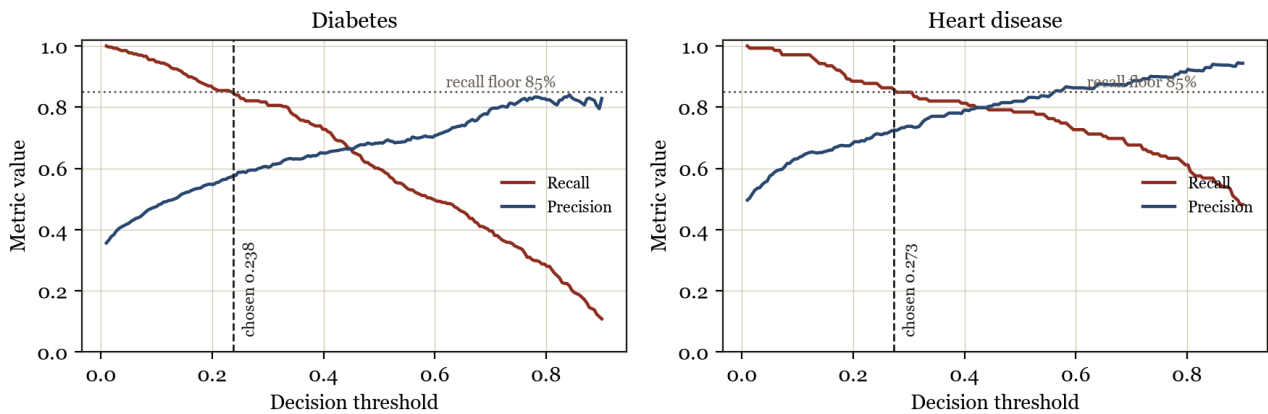


Figure 10: Precision and recall as the threshold varies. The chosen operating point is the best-precision threshold that keeps recall above the 85% floor — a stated rule, not a tuned constant.

7.3 Calibration

Figure 11 groups out-of-fold predictions into deciles and compares mean predicted risk with the observed disease rate in each group. Both diseases track the diagonal closely, meaning the probabilities shown to users are honest: patients told "about 30% risk" develop the disease roughly 30% of the time. This matters because the entire interface — tiers, bands, what-if deltas — is denominated in these probabilities.

Calibration: predicted risk vs what actually happened (out-of-fold deciles)

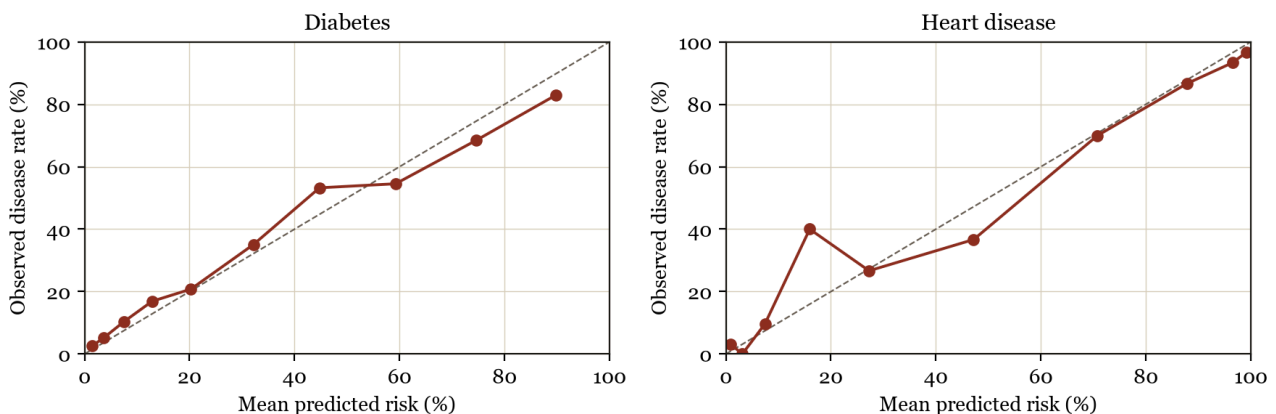


Figure 11: Reliability diagrams (out-of-fold deciles). Points near the diagonal indicate honest probabilities.

7.4 Is the uncertainty informative?

The central claim of this project is that the confidence band is not decoration. To test it, out-of-fold patients were sorted into quartiles by the width of their 90% band, and the classification error rate was measured within each quartile (Figure 13). The effect is large and monotonic: for diabetes, the error rate rises from **6%** in the narrowest-band quartile to **43%** in the widest; for heart disease, from **4%** to **46%**. When the 200 models agree, they are almost always right; when they disagree, the system says so — and it is precisely then that it errs. A user who trusts narrow bands and treats wide bands as "get a proper test" is served far better than by any single number.

The system knows when it does not know: wider bands, more mistakes

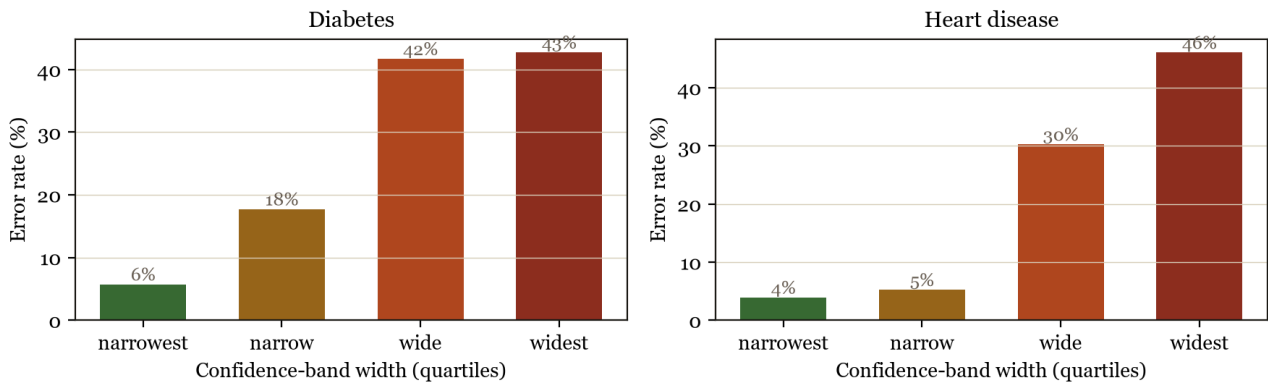


Figure 12: Error rate by confidence-band width (out-of-fold quartiles). Wide bands concentrate the mistakes: the system knows when it does not know.

Figure 12 shows what this looks like for individual patients: a healthy profile yields a needle-thin distribution (band 1–2%), while a high-risk profile yields a visibly wide one (65–96%) — honest uncertainty a clinician can read at a glance.

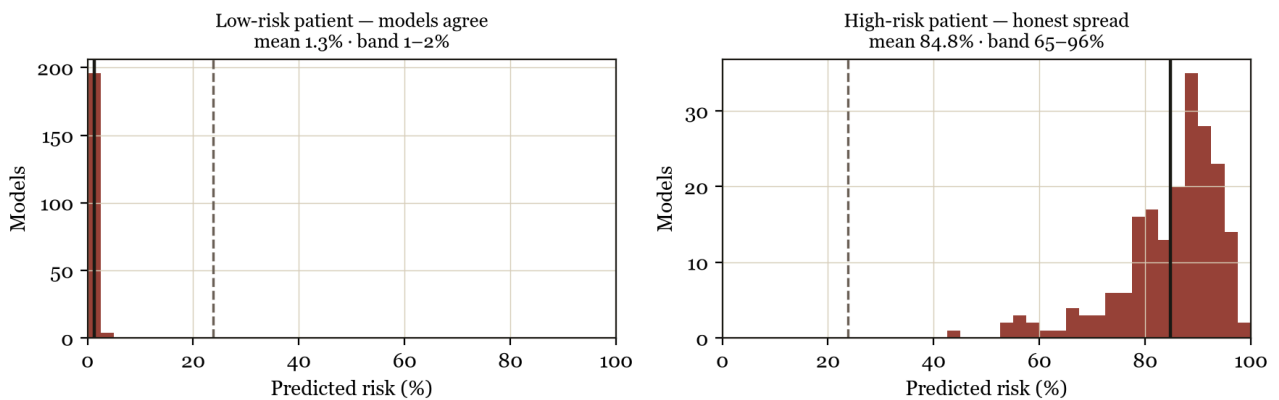


Figure 13: Per-patient risk distributions from the 200 members: agreement for a clear case (left), honest spread for an ambiguous one (right). Solid line: mean; dashed: alert threshold.

7.5 Per-patient explanations

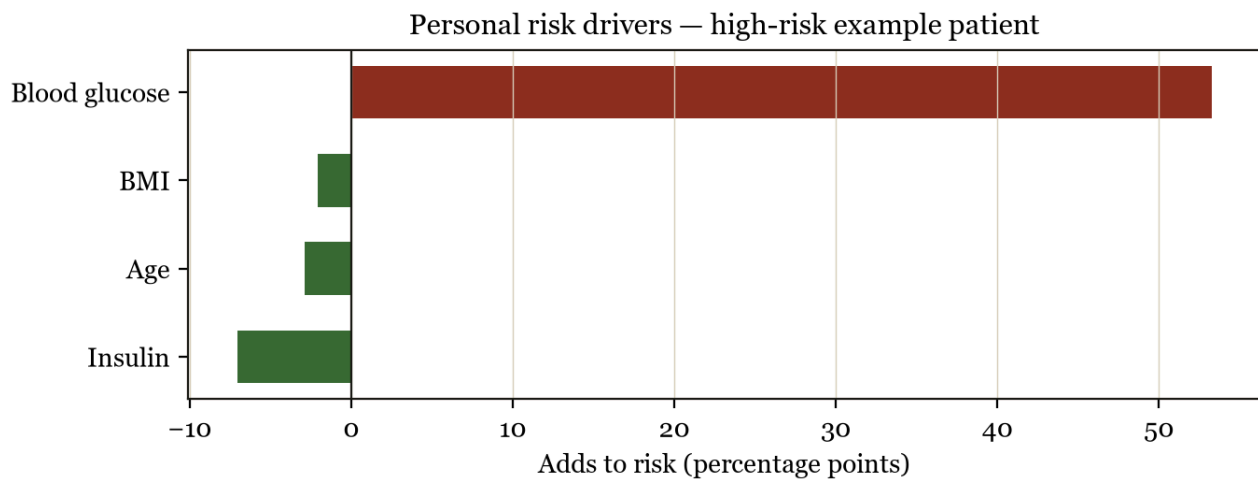


Figure 14: Personal risk drivers for the high-risk example patient (median substitution). Glucose dominates, matching clinical intuition.

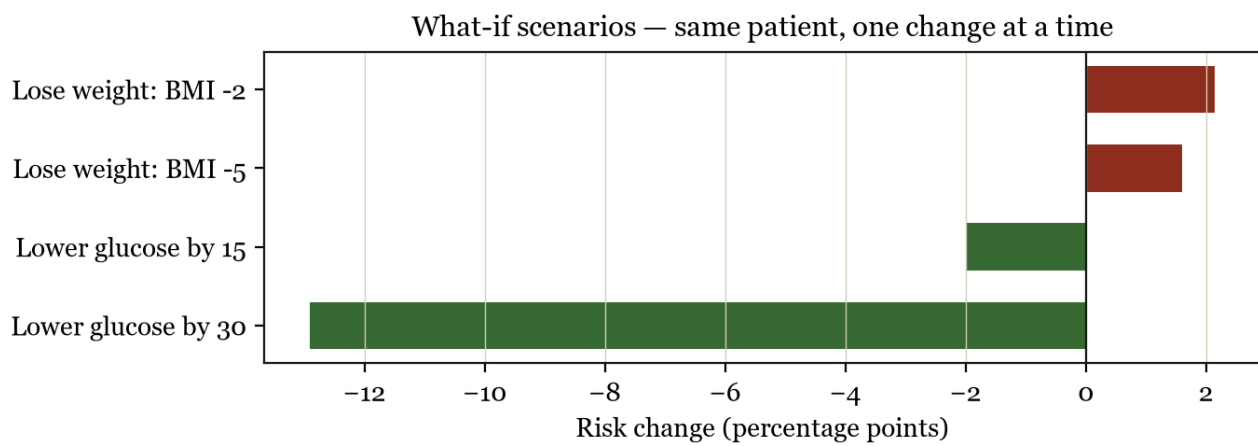


Figure 15: What-if simulation for the same patient: each bar re-runs all 200 models with one value changed.

8. The Dashboard

The delivery layer is a Streamlit application styled as an interactive lab report — paper ground, ink text, a single oxblood accent carrying all risk semantics — and deployed publicly at **healthpredictor092005.streamlit.app**. Its defining structural feature is a *gated flow*: on arrival the user sees an intake explainer and a form (Figure 16); no prediction exists until "Run assessment" is pressed, and the resulting report always opens with a summary strip naming exactly whose values it describes. This eliminates the confusing phantom-prediction state common in dashboard templates, where output for a default patient is visible before any input has been given.

Patient intake

Assessment for

☒ Diabetes ☐ Heart disease

Glucose (2h OGTT, mg/dL)

120 - +

BMI

30.0 - +

Age (years)

33 - +

Blood pressure (diastolic, mm Hg)

70 - +

Pregnancies

2 - +

Family history score (pedigree) ⓘ

0.40 - +

Insulin (2h serum, mu U/ml)

100 - +

☐ Insulin unknown

Skin thickness (mm)

Deploy ⋮

Early Disease Prediction — Assessment Report

Disease risk before symptoms — diabetes and heart disease. Two hundred models vote; the fifty most similar real patients give an independent second opinion.

No assessment has been run yet. This instrument produces a risk report only when you ask it to.

- 1 Choose a disease and enter the patient's data**
Pick diabetes or heart disease, then fill the intake form on the left. Values you don't know can be marked unknown — the model imputes the study median.
- 2 Run the assessment**
All 200 models score the patient; the 50 most similar real patients are found for a second opinion.
- 3 Read the report**
Risk with uncertainty, what drives it, what would change it, and a preventive plan — each in its own section.

Figure 16: First-run state: the instrument explains itself; nothing has been predicted.

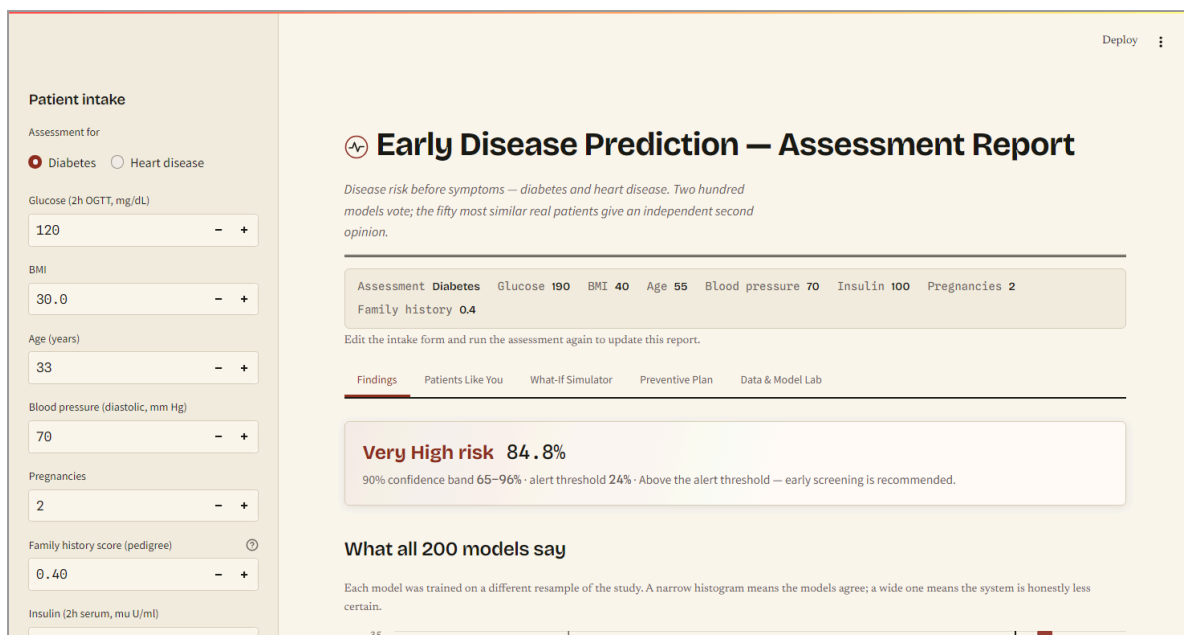


Figure 17: Findings — diabetes example patient: tier, calibrated risk, 90% band, and the distribution of all 200 model opinions.

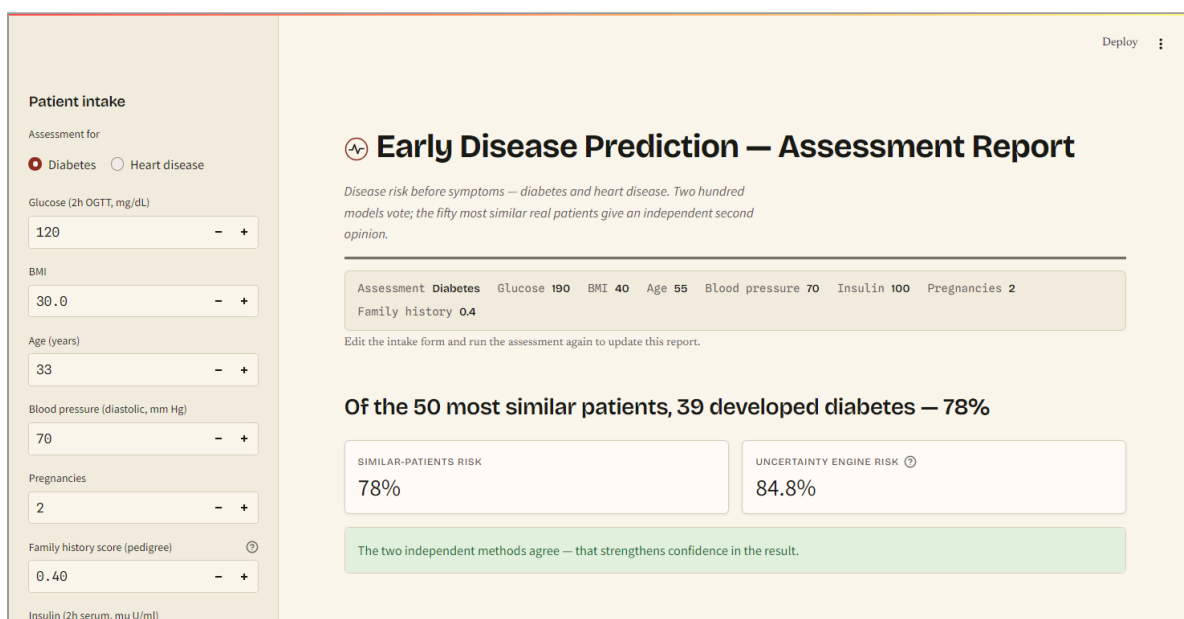


Figure 18: Patients Like You: the 50 most similar real patients plotted in glucose–BMI space, with the model-free second opinion and the agreement check.

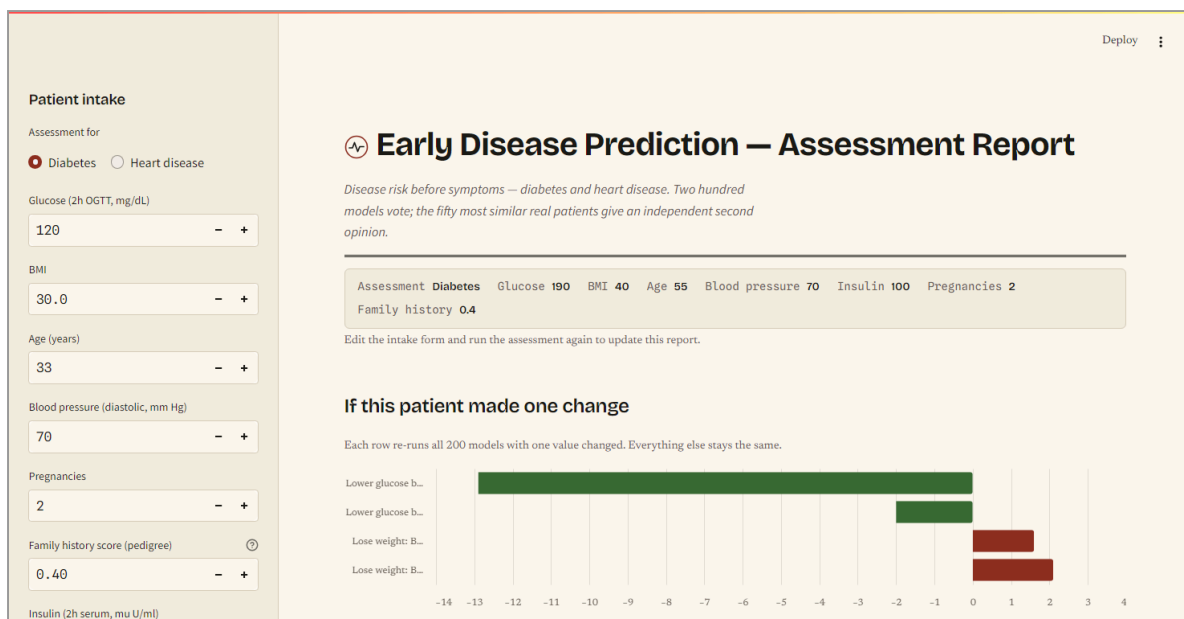


Figure 19: What-If Simulator: achievable changes re-scored by the full ensemble.

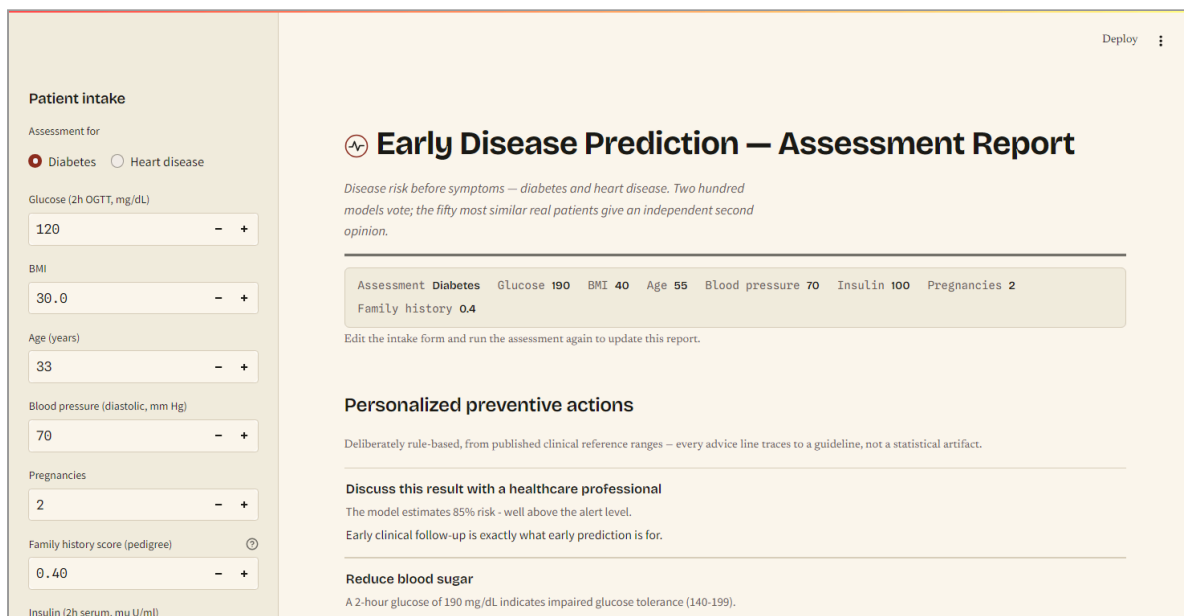


Figure 20: Preventive Plan: rule-based actions citing the patient's own values and the guideline ranges that triggered them.

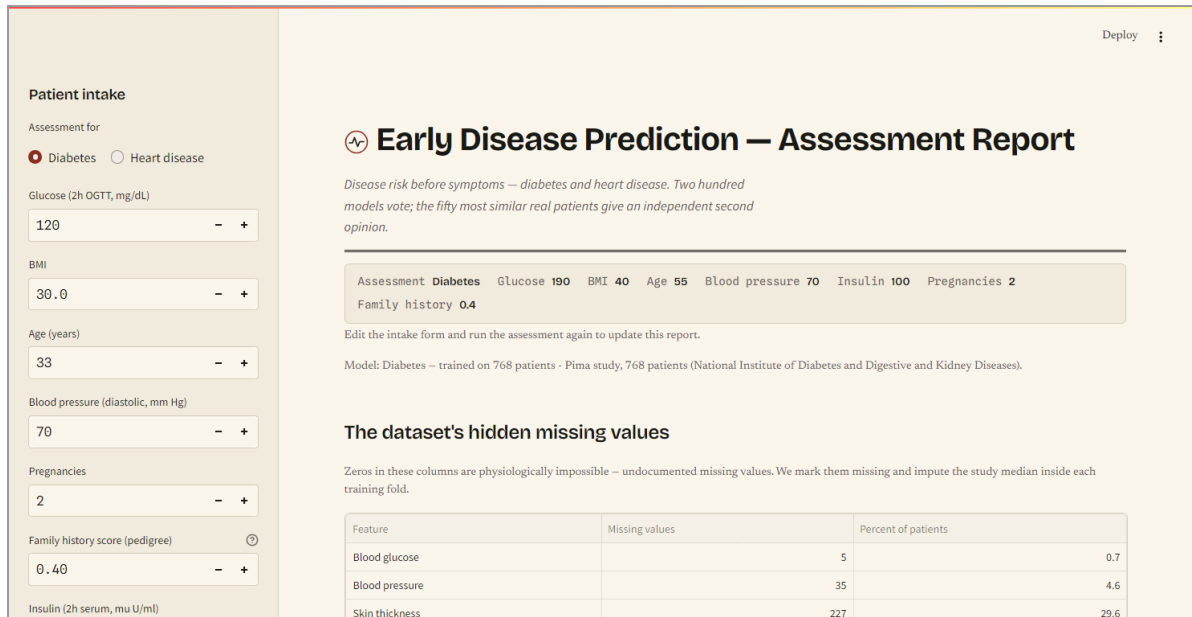


Figure 21: Data & Model Lab: the transparency page — hidden-missing-data audit, out-of-fold metrics, model comparison, threshold rule and calibration.

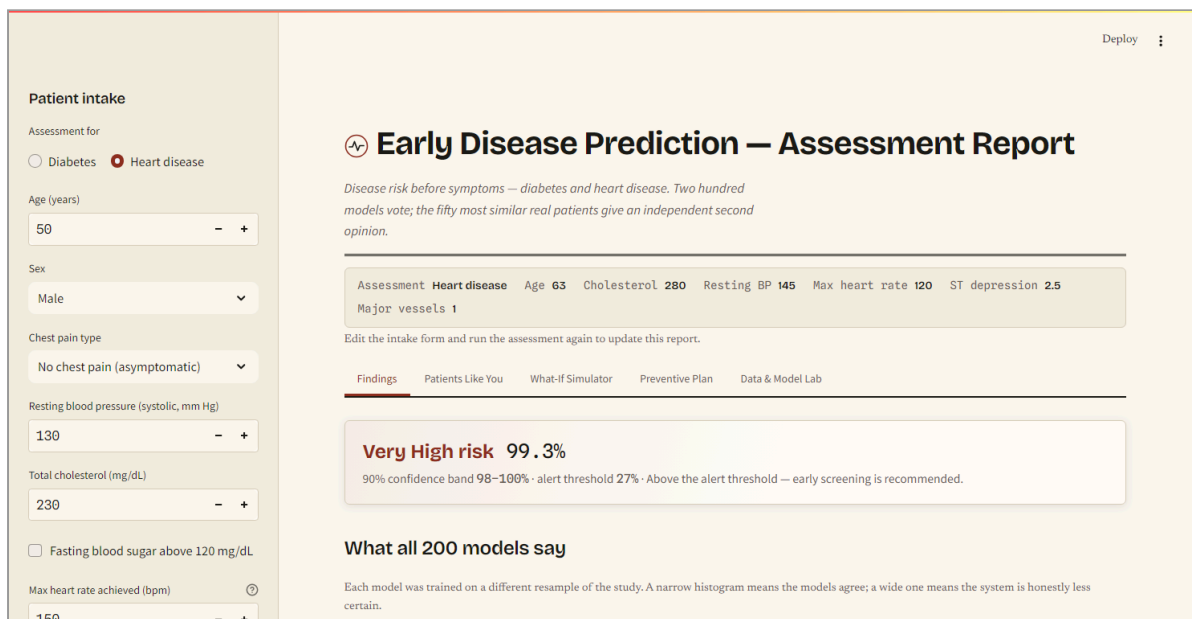


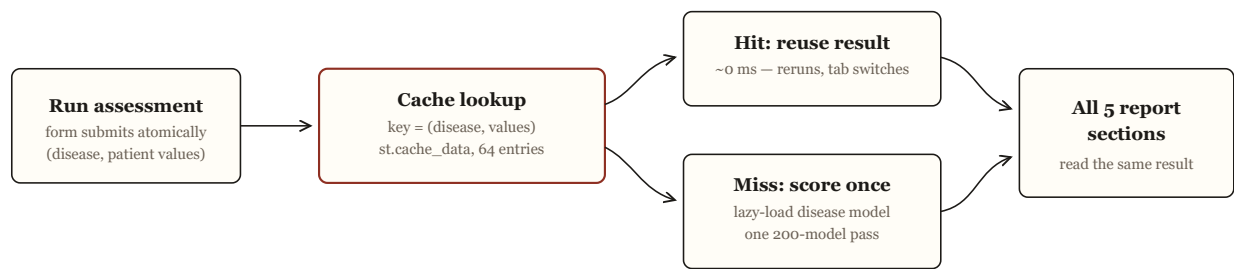
Figure 22: The same interface serving the heart-disease model — one engine, two diseases.

9. Engineering and Deployment

The codebase separates a disease-agnostic engine (`src/edp/`) from per-disease plug-in configurations (`src/edp/diseases/`) and the interface (`ui/`). A 15-test pytest suite guards the data repair, decision logic, explanation builders and both engines. Training is a single reproducible script; every figure in this report was generated by a committed script from the trained artifacts.

Serving performance was engineered deliberately (Figure 18): each patient is scored *exactly once* — the 200-member pass is memoized by (disease, patient values) and all five report sections read the same cached result — and a disease's 36 MB model file is loaded lazily, only when that disease is first assessed.

Measured on the free-tier cloud deployment, interaction latency fell from roughly 22 seconds to 2–4 seconds after this change; only the first assessment after a cold start pays the model-loading cost.



*Each patient is scored exactly once; the interface only ever re-reads the memoized result.
Measured on the live deployment: 22 s per interaction before → 2–4 s after.*

Figure 23: Serving architecture: score-once caching with lazy model loading.

10. Limitations

The system is an educational prototype and its results must be read accordingly:

- **Small, dated cohorts.** 768 and 303 patients, collected decades ago from specific populations (Pima women aged 21+; Cleveland angiography patients). Absolute risk levels do not transfer to other populations without recalibration.
- **No external validation.** All evaluation is cross-validation within each dataset; a genuinely new cohort could expose distribution shift.
- **Correlation, not causation.** The what-if simulator answers "what would the model predict if this value were different", which is not identical to the causal effect of an intervention on the patient.
- **Bootstrap uncertainty is sampling uncertainty.** The band reflects disagreement between models trained on resamples; it cannot capture uncertainty about factors the data never measured.
- **Not clinically validated.** The application states in-product that it is an educational tool and that decisions belong with healthcare professionals.

11. Conclusion and Future Work

This project set out to show that an early disease-prediction system can be honest about what it knows. The delivered system predicts diabetes and heart-disease risk with out-of-fold ROC AUC of 83.5% and 90.0% at $\geq 85\%$ recall by construction; its probabilities are calibrated; and — the central result — its per-patient confidence bands are demonstrably informative, with error rates roughly seven to ten times higher among the widest bands than the narrowest. A model-free second opinion, jargon-free explanations, a what-if simulator and guideline-based preventive plans convert the score into something a person can act on, and the disease-registry architecture makes the whole construction extensible at the cost of one configuration file per new disease.

Natural next steps: adding a third disease (chronic kidney disease has a suitable UCI dataset) to further demonstrate the registry; external validation on a modern cohort such as NHANES; conformal prediction as a complementary uncertainty method with formal coverage guarantees; and a longitudinal mode that tracks a patient's risk trajectory across repeated assessments.

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